

ODUOLA, IDRIS ADEWALE

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PERSONAL STATEMENT

As a graduate in Mathematics with interests in computational and machine learning methods for PDE-based problems in physics and engineering, I aspire to pursue a career in research and teaching.

EDUCATION

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia
Master of Science in Mathematics (CGPA: 3.78/4.0)

Aug 2022 – Jun 2024

University of Ilorin, Ilorin, Nigeria
Bachelor of Science in Mathematics (CGPA: 4.91/5.0)

Oct 2017 – Oct 2021

RESEARCH EXPERIENCE

Graduate Research Assistant
King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

May 2023 - Present

- Applied machine learning to solve Partial Differential Equations (PDEs).
- Developed PINN algorithms for Quantum Eigenvalue Problems.
- Developing structure-preserving machine learning surrogates for multi-phase flows in porous media.

Undergraduate Research Assistant
University of Ilorin, Ilorin, Nigeria

Oct 2019 - May 2021

- Built a deterministic model to study declining interest in university-level mathematics.
- Analyzed equilibrium states using Jacobian matrix and solved the model with the Differential Transform Method (DTM).

PROJECTS

See the Projects section of my webpage for illustrative examples of recent work. ([click here](#))

Full Waveform Inversion as a Gradient Flow Problem

Ongoing

- Developing a physics-guided machine learning model to solve full-waveform inversion problems.
- Currently, the model will utilize separate networks for the forward and inversion processes, with an evolutionary neural network approach.
- The forward model has a positional-encoding mechanism to ensure sequence position of seismic sources are correctly identified. *Initial progress can be found [here](#)*

Hands-on Scientific Machine Learning eBook

Ongoing

- Creating an online interactive eBook for recent advances in Scientific machine learning, in collaboration with Xiaoyu Xie ([personal website](#)). *Project progress can be found [here](#).*

Physics Informed Neural Network (PINN) for Quantum Eigenvalue Problems

2024 - 2025

- Developed PINN algorithms to approximate the wavefunctions of quantum wells, and Rayleigh quotient to compute the corresponding energy levels. Two strategies were considered:
 - **Deflation Approach:** Eigenstates are learned sequentially using transfer learning and a deflation method. ([Check codes here](#))
 - **Simultaneous Multi-state Approach:** A self-attention mechanism is incorporated as an orthogonalization constraint, allowing arbitrary number of eigenstates to be returned in a single forward pass. ([Check codes here](#))

Numerical Solution of Lorenz Equations: Application in Fluid Dynamics and Atmospheric Convection
KFUPM Math 571 Course

2023

- Employed numerical techniques to investigate the solution of the Lorenz equations.

- Captured the performance of different techniques in solving the Lorenz equations and proposed the best technique.

Optimizing Neural Networks for Predicting Titanic Survivorship

2022

Kaggle

- Built a neural network exclusively using NumPy, matplotlib, pandas, and sklearn preprocessing tool, for the Kaggle Titanic competition. Major optimization techniques were employed to enhance model performance. ([Check mini-project here](#))

University of Michigan (Coursera)

2021

Web Design for Everybody Capstone

- Built a responsive, professional-quality travel blog portfolio, including an enhanced variant with plugins (e.g., Bootstrap Carousel), ensuring accessibility and validation compliance. Links: [Core version](#) | [Enhanced version](#)

CONFERENCES, SEMINARS AND WORKSHOPS

Finite Volume Schemes for Diffusion–Convection Equations in Porous Media (Workshop)

Sep 2025

KFUPM Department of Mathematics

- Participated in a 3-day workshop on finite volume methods for diffusion–convection equations in porous media, taught by Prof. Mazen Saad (École Centrale de Nantes).

Data to Decisions in the E&P – Bridging the Gap with Machine Learning (Workshop)

Feb 2025

KFUPM College of Petroleum Engineering and Geosciences

- Attended a short course on data-driven decision making and predictive modeling with AI/ML in exploration & production (E&P).

MathConnect 2024: International Conference of Mathematics and its Applications

Dec 2024

KFUPM

- Attended the 2024 MathConnect conference featuring recent advances across various domains of Mathematics.

First KFUPM Interdisciplinary Graduate Student Conference

May 2024

KFUPM

- Presented my research work with the title, “*Can Neural Networks learn Eigenpairs?*”
- Awarded the 3rd research presentation award with monetary prize for the talk.

Physics Informed Neural Networks (PINNs) for Quantum Eigenvalue Problems

Nov 2023

KFUPM Mathematics Department Seminar

- Gave a talk on PINN and its implementation in solving Quantum Eigenvalue problems, highlighting advantages over traditional numerical methods.

AWARDS AND HONORS

3rd Place Research Presentation Award

2024

First KFUPM Interdisciplinary Graduate Student Conference

- Received prize and cash award (2000 SAR) for research presentation

Graduate Scholarship (MSc Applied Mathematics)

2022

KFUPM

- Full scholarship awarded for master’s studies.

Combined Master/Doctoral Research & Teaching Scholarship

2022

Khalifa University

- Full scholarship in Computational Data Science (offer declined due to visa restrictions)

Federal Scholarship Board (FSB) National Awards

2020

Federal Ministry of Education

- Awarded to top-performing students nationwide following competitive aptitude testing.

NAMSSN Award of Academic Excellence

2019

University of Ilorin

- Recognized for achieving a CGPA of 4.98/5.0 (2018–2019 session).

CERTIFICATIONS

Deep Neural Networks with Pytorch – IBM (Coursera)

Aug 2023

- Implemented linear/logistic regression, deep neural networks, CNNs, Softmax, dropout using PyTorch.

Machine Learning Specialization – DeepLearning.AI (Coursera)

Jun 2022

- Covered supervised ML, regression/classification, and advanced learning algorithms.

Developing the SIR Model – Imperial College London (Coursera)

Aug 2021

- Studied compartmental models, interpreting and representing rates, durations, and proportions.

Research Writing in the Sciences – INASP

May 2021

- Learned to review literature, apply ethical research practices, write scientific manuscripts, and publish in peer-reviewed journals. *Credential ID: KnCTb3GdiC*

For a complete list of certifications, please refer to the [Experience section](#) of my webpage.

TEACHING EXPERIENCE

Teaching Assistant

Sep 2022-Present

KFUPM

- Taught MATLAB and Python lab sessions for MATH 101/102.
- Proctored and graded quizzes, exams, and assignments.

Mathematics Tutor

Jun 2021- Sep 2021

Lumex Academy, Ilorin, Nigeria

- Taught calculus and differential equations for JUPEB students.
- Designed assessments to strengthen comprehension, improving student performance.

Mathematics Tutor

Jan 2021-Jun 2021

Benchmark Academy, Ilorin, Nigeria

- Taught pre-calculus, calculus, differential equations, and geometry to prospective undergraduates.

SKILLS

Computer

Python, C, SQL, MATLAB, R, FORTRAN, LaTeX, HTML5, CSS3, JavaScript

Personal

Creative Writing, Academic Writing, Problem Solving, Teaching

INTERESTS

Scientific Machine Learning, Data-driven reduced-order modeling, Inverse Problems, Computational Fluid Dynamics (CFD).